

The Unit Circle and Exact Values



Exact values

1 Write the exact value of each expression:

a $\sin \frac{\pi}{6}$

b $\cos \frac{3\pi}{4}$

c $\tan \frac{5\pi}{6}$

d $\sin \frac{3\pi}{2}$

2 Write the exact value of each expression:

a $\cos \left(-\frac{\pi}{3}\right)$

b $\sin \frac{7\pi}{4}$

c $\tan \pi$

d $\cos \frac{11\pi}{6}$

Points on the unit circle

3 The terminal side of angle θ in standard position passes through the point $P\left(-\frac{3}{5}, \frac{4}{5}\right)$ on the unit circle. Find:

a $\sin \theta$

b $\cos \theta$

c $\tan \theta$

d the quadrant containing θ

4 State the reference angle for each angle:

a $\frac{5\pi}{6}$

b $\frac{4\pi}{3}$

c $-\frac{\pi}{4}$

d $\frac{7\pi}{2}$

5 Identify the quadrant(s) in which each pair of conditions holds:

a $\sin \theta < 0$ and $\cos \theta > 0$

b $\tan \theta > 0$ and $\sin \theta < 0$

6 Given that $\cos \theta = -\frac{5}{13}$ and θ is in Quadrant III, find $\sin \theta$ and $\tan \theta$ exactly.

Even/odd properties and periodicity

7 Use the fact that cosine is even and sine is odd to simplify each expression:

a $\sin(-\theta) + \sin \theta$

b $\cos(-\theta) - \cos \theta$

c $\frac{\sin(-\theta)}{\cos(-\theta)}$

8 Use periodicity to find the exact value of each expression:

a $\sin \frac{13\pi}{6}$

b $\cos \frac{17\pi}{4}$

c $\tan \frac{9\pi}{4}$

Solving basic trigonometric equations on the unit circle

9 Find all solutions to $\sin \theta = -\frac{1}{2}$ in the interval $[0, 2\pi)$.

10 Find all solutions to $2\cos \theta + \sqrt{3} = 0$ in the interval $[0, 2\pi)$.

Building the unit circle from special triangles

11 Using the 30-60-90 triangle, derive the exact coordinates of the point on the unit circle at $\theta = \frac{\pi}{3}$.

12 State the coordinates of all four points on the unit circle where the terminal side lies along an axis ($\theta = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}$).

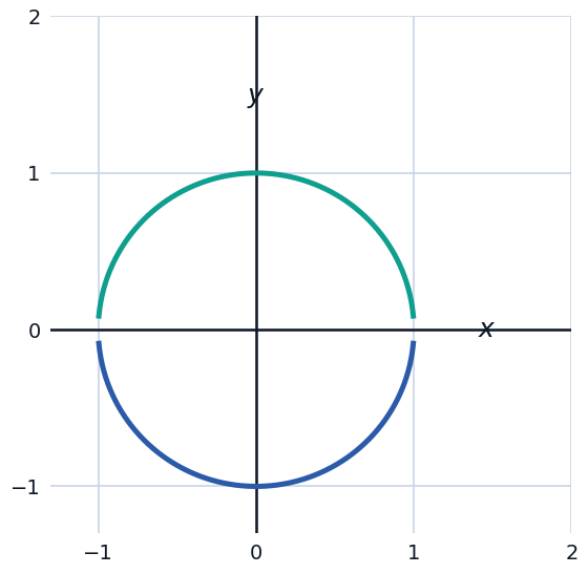
Using the Pythagorean identity with unit circle points

13 A point on the unit circle has x-coordinate $\frac{7}{25}$ and lies in Quadrant IV. Find its y-coordinate using $x^2 + y^2 = 1$.

14 Given $\sin \theta = 0.6$ and θ in Quadrant II, find $\cos \theta$ and $\tan \theta$ exactly (using the fact $0.6 = \frac{3}{5}$).

The unit circle picture

- 15 The unit circle $x^2 + y^2 = 1$ is shown. Use it to estimate $\cos\left(\frac{2\pi}{3}\right)$ and $\sin\left(\frac{2\pi}{3}\right)$ visually, then state the exact values.



All six trig functions from a unit circle point

- 16 A point on the unit circle has coordinates $\left(-\frac{5}{13}, \frac{12}{13}\right)$. Find all six trigonometric function values for the corresponding angle θ .

Solving for the angle given exact coordinates

- 17 Find all values of θ in $[0, 2\pi)$ for which the unit circle point is exactly $\left(-\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right)$.